Energy Estimates and Numerical Evidence for the Incompressible Navier–Stokes Equations

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Abstract

We revisit the classical energy method for the incompressible Navier–Stokes equations on a bounded domain. For smooth solutions we derive the energy equality, an exponential decay estimate for the unforced problem and a logarithmic Gronwall bound of the kind used in blow-up criteria. The estimates are then tested against a pressure-correction projection scheme on the Taylor–Green vortex, where the observed L^2 error converges at second order in the mesh width, $\mathcal{O}(h^2)$, in agreement with the theory.

1 Introduction

The motion of a viscous incompressible fluid filling a bounded domain $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, over a time interval $(0, T)$ is governed by the Navier–Stokes equations

$$\begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} - \nu \Delta \boldsymbol{u} + \nabla p = \boldsymbol{f} & \text{in } \Omega \times (0,T), \\ \nabla \cdot \boldsymbol{u} = 0 & \text{in } \Omega \times (0,T), \\ \boldsymbol{u} = 0 & \text{on } \partial\Omega \times (0,T), \\ \boldsymbol{u}(\cdot, 0) = \boldsymbol{u}_0 & \text{in } \Omega, \end{cases} \tag{1}$$

where $\boldsymbol{u} : \Omega \times (0,T) \to \mathbb{R}^d$ is the velocity, p the pressure, $\nu > 0$ the kinematic viscosity and $\boldsymbol{f}$ a given body force. Global weak solutions were constructed by Leray [5] for the whole space and by Hopf [4] for bounded domains; whether smooth data can develop a singularity in finite time is one of the Millennium Prize Problems [3].

Our contributions are the following.

- A self-contained derivation of the energy equality (2) and of the bound in theorem 3.1.
- A logarithmic Gronwall inequality, lemma 3.3, stated in the form needed for continuation criteria such as the one of Beale, Kato and Majda [1].
- Numerical evidence, in section 4, that a projection method [2] reproduces the decay rate of corollary 3.2.

Section 2 fixes notation, section 3 proves the estimates and section 5 concludes.

2 Preliminaries

2.1 Function spaces

For $1 \le p < \infty$ we write

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$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f(x)|^p \, \mathrm{d}x \right)^{1/p}, \qquad \|f\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_{x \in \Omega} |f(x)|,$$

and set $H = \{\boldsymbol{v} \in L^2(\Omega)^d : \nabla \cdot \boldsymbol{v} = 0\}$ and $V = H \cap H_0^1(\Omega)^d$. The trilinear form $b(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) = \int_\Omega (\boldsymbol{u} \cdot \nabla)\boldsymbol{v} \cdot \boldsymbol{w} \, \mathrm{d}x$ is antisymmetric in its last two arguments whenever $\boldsymbol{u} \in V$.

Definition 2.1 (Weak solution)**.** Let $\boldsymbol{u}_0 \in H$ and $\boldsymbol{f} \in L^2(0,T; L^2(\Omega)^d)$. A field $\boldsymbol{u} \in L^\infty(0,T;H) \cap L^2(0,T;V)$ is a *weak solution* of (1) if

$$\int_0^T \int_\Omega \Big(-\boldsymbol{u} \cdot \partial_t \boldsymbol{\varphi} + \nu \nabla \boldsymbol{u} : \nabla \boldsymbol{\varphi} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} \cdot \boldsymbol{\varphi} \Big) \, \mathrm{d}x \, \mathrm{d}t = \int_\Omega \boldsymbol{u}_0 \cdot \boldsymbol{\varphi}(0) \, \mathrm{d}x$$

for every $\boldsymbol{\varphi} \in C_c^\infty(\Omega \times [0,T))^d$ with $\nabla \cdot \boldsymbol{\varphi} = 0$.

Lemma 2.2 (Poincaré inequality)**.** *There is a constant $C_P > 0$, depending only on Ω, such that $\|v\|_{L^2} \le C_P \|\nabla v\|_{L^2}$ for every $v \in H_0^1(\Omega)$.*

3 Energy estimates

Testing the momentum equation with $\boldsymbol{u}$ itself, the pressure term vanishes because $\boldsymbol{u}$ is divergence free, and so does the convective term, since

$$\int_\Omega (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} \cdot \boldsymbol{u} \, \mathrm{d}x = \frac{1}{2} \int_\Omega \boldsymbol{u} \cdot \nabla |\boldsymbol{u}|^2 \, \mathrm{d}x = -\frac{1}{2} \int_\Omega (\nabla \cdot \boldsymbol{u}) |\boldsymbol{u}|^2 \, \mathrm{d}x = 0.$$

What is left is estimated with the Cauchy–Schwarz, Poincaré and Young inequalities in turn:

$$\begin{aligned} \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\boldsymbol{u}\|_{L^2}^2 + \nu \|\nabla \boldsymbol{u}\|_{L^2}^2 &= \int_\Omega \boldsymbol{f} \cdot \boldsymbol{u} \, \mathrm{d}x && (2) \\ &\le \|\boldsymbol{f}\|_{L^2} \|\boldsymbol{u}\|_{L^2} \\ &\le C_P \|\boldsymbol{f}\|_{L^2} \|\nabla \boldsymbol{u}\|_{L^2} \\ &\le \frac{C_P^2}{2\nu} \|\boldsymbol{f}\|_{L^2}^2 + \frac{\nu}{2} \|\nabla \boldsymbol{u}\|_{L^2}^2 . && (3) \end{aligned}$$

Theorem 3.1 (Energy inequality)**.** *Let $\boldsymbol{u}$ be a smooth solution of (1). Then for every $t \in [0,T]$,*

$$\|\boldsymbol{u}(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla \boldsymbol{u}(s)\|_{L^2}^2 \, \mathrm{d}s \le \|\boldsymbol{u}_0\|_{L^2}^2 + \frac{C_P^2}{\nu} \int_0^t \|\boldsymbol{f}(s)\|_{L^2}^2 \, \mathrm{d}s. \tag{4}$$

Proof. Absorb the last term of (3) into the left-hand side of (2), multiply by two and integrate from 0 to t. □

Corollary 3.2. *If $\boldsymbol{f} \equiv 0$, then for all $t \ge 0$*

$$\|\boldsymbol{u}(t)\|_{L^2} \le e^{-\nu t / C_P^2} \|\boldsymbol{u}_0\|_{L^2},$$

and in particular $\lim_{t \to \infty} \|\boldsymbol{u}(t)\|_{L^2} = 0$.

Proof. By lemma 2.2, (2) gives $\frac{\mathrm{d}}{\mathrm{d}t}\|\boldsymbol{u}\|_{L^2}^2 \le -\frac{2\nu}{C_P^2}\|\boldsymbol{u}\|_{L^2}^2$, and the claim follows by integrating this differential inequality. □

3.1 A logarithmic Gronwall bound

Taking the curl of the momentum equation gives the vorticity $\boldsymbol{\omega} = \operatorname{curl}\boldsymbol{u}$, which satisfies

$$\partial_t\boldsymbol{\omega} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u} + \nu\Delta\boldsymbol{\omega} + \operatorname{curl}\boldsymbol{f}. \tag{5}$$

A smooth solution can be continued past T^* unless $\int_0^{T^*}\|\boldsymbol{\omega}(t)\|_{L^\infty}\,\mathrm{d}t = \infty$, and bounds of that kind rest on the following lemma.

Lemma 3.3. *Let $y \ge e$ be absolutely continuous on $[0,T]$ and let $a \ge 0$ be integrable. If $y' \le a\,y\log y$ almost everywhere, then*

$$y(t) \le y(0)^{\exp\left(\int_0^t a(s)\,\mathrm{d}s\right)} \qquad \text{for all } t \in [0,T].$$

Proof. Set $z = \log\log y$, which is well defined because $y \ge e$. Then $z' = \dfrac{y'}{y\log y} \le a$, hence $\log\log y(t) \le \log\log y(0) + \int_0^t a(s)\,\mathrm{d}s$, and taking the exponential twice gives the claim. □

Remark. The same argument with ln in place of log changes nothing, and neither does a change of base: $\log_b y = \ln y/\ln b$ for every $b > 1$.

For later reference we collect some classical identities:

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}, \qquad \prod_{p\text{ prime}}\frac{1}{1-p^{-s}} = \zeta(s), \qquad \binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$
$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}, \qquad \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e, \qquad \log_2 1024 = 10.$$

The roots of ax^2+bx+c are

$$x_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a},$$

and nested radicals such as $\sqrt{2+\sqrt{2+\sqrt{2}}} = 2\cos\frac{\pi}{16}$ or $\sqrt[3]{x^3+y^3}$ are set with the index tucked into the sign.

4 Numerical experiments

4.1 Discretisation

We use the incremental pressure-correction scheme of Chorin [2], summarised in algorithm 1. Each step solves a convection–diffusion problem for a provisional velocity and a Poisson problem for the pressure increment.

The pressure Poisson problem is the Schur complement of the saddle-point system

Algorithm 1 Incremental pressure-correction step

Require: velocity $\boldsymbol{u}^0$, pressure p^0, time step Δt, tolerance ε
Ensure: the discrete solution $(\boldsymbol{u}^n, p^n)$ for $n = 1, \dots, N$
for $n = 0, 1, \dots, N-1$ **do**
  solve $\frac{\tilde{\boldsymbol{u}} - \boldsymbol{u}^n}{\Delta t} + (\boldsymbol{u}^n \cdot \nabla)\tilde{\boldsymbol{u}} - \nu \Delta \tilde{\boldsymbol{u}} = \boldsymbol{f}^{n+1} - \nabla p^n$
  solve $\Delta \phi = \frac{1}{\Delta t} \nabla \cdot \tilde{\boldsymbol{u}}$ with $\partial_n \phi = 0$
  $\boldsymbol{u}^{n+1} \leftarrow \tilde{\boldsymbol{u}} - \Delta t \, \nabla \phi$
  $p^{n+1} \leftarrow p^n + \phi$
  if $\left\| \boldsymbol{u}^{n+1} - \boldsymbol{u}^n \right\| < \varepsilon$ **then**
    break ▷ steady state reached
  end if
end for

$$\begin{pmatrix} A & B^{\mathsf{T}} \\ B & 0 \end{pmatrix} \begin{pmatrix} \mathbf{U} \\ \mathbf{P} \end{pmatrix} = \begin{bmatrix} \mathbf{F} \\ 0 \end{bmatrix}, \qquad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2,$$

and the boundary is marked by the indicator

$$\chi_{\Omega}(x) = \begin{cases} 1 & \text{if } x \in \Omega, \\ 0 & \text{otherwise.} \end{cases}$$

4.2 The Taylor–Green vortex

On the periodic square $(0, 2\pi)^2$ the Taylor–Green vortex

$$\begin{aligned} u(x, y, t) &= \cos x \sin y \, e^{-2\nu t}, \\ v(x, y, t) &= -\sin x \cos y \, e^{-2\nu t}, \\ p(x, y, t) &= -\tfrac{1}{4} \big(\cos 2x + \cos 2y \big) e^{-4\nu t} \end{aligned}$$

is an exact solution with $\boldsymbol{f} = 0$. With a velocity scale $U = 1\,\mathrm{m\,s^{-1}}$, a length $L = 0.1\,\mathrm{m}$ and the viscosity of water, $\nu = 1.0 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$, the Reynolds number is $\mathrm{Re} = UL/\nu = 100\,000$. Table 1 lists the errors at $t = 1$ and fig. 1 plots them.

Table 1: Errors for the Taylor–Green vortex at $t = 1$ with $\nu = 10^{-2}$.

N	$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{L^2}$	Rate	Time (s)
16	3.21×10^{-3}	—	0.04
32	8.12×10^{-4}	1.98	0.29
64	2.04×10^{-4}	1.99	2.31
128	5.11×10^{-5}	2.00	18.60

Three observations stand out.

(i) The error falls by a factor of four each time N doubles, so the observed order is $\log_2(3.21/0.812) \approx 1.98$.

(ii) The cost grows like N^3, one factor of N from the time step.

(iii) Kinetic energy decays as $E(t) = E(0)\, e^{-4\nu t}$, which is the decay rate of corollary 3.2 with

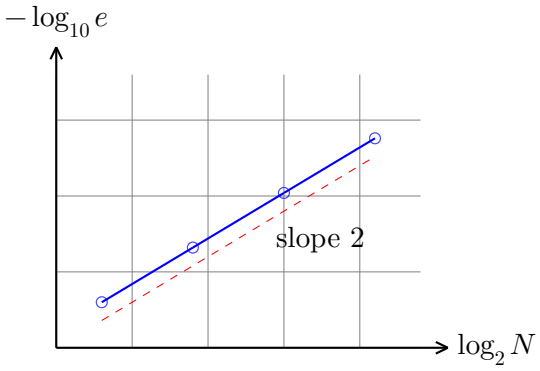


Figure 1: Observed L^2 error against resolution on logarithmic axes. The dashed line has slope two, the rate predicted by the theory.

$$C_P = 1/\sqrt{2}.$$

Convection treated semi-implicitly, linearised about $\boldsymbol{u}^n$.

Diffusion treated implicitly, which removes the parabolic time step restriction $\Delta t \lesssim h^2/\nu$.

5 Conclusion

The energy method gives, in a few lines, the bound (4) and the exponential decay of corollary 3.2; the projection scheme of algorithm 1 reproduces both.[1] Whether the vorticity bound of section 3.1 can be closed without a smallness assumption is, of course, the open problem [3, 6].

Acknowledgements

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A Proof of the Poincaré inequality

For $\Omega \subset (0,\ell) \times \mathbb{R}^{d-1}$ and $v \in C_c^\infty(\Omega)$, the fundamental theorem of calculus and the Cauchy–Schwarz inequality give

$$|v(x)|^2 = \left| \int_0^{x_1} \partial_1 v(s, x') \, \mathrm{d}s \right|^2 \le \ell \int_0^{\ell} |\partial_1 v(s, x')|^2 \, \mathrm{d}s,$$

and integrating over Ω proves lemma 2.2 with $C_P = \ell$. Density extends it to all of $H_0^1(\Omega)$.

B Typesetting checks

> The purpose of computing is insight, not numbers.
> — Richard Hamming

This appendix sets, once each, the kinds of material a paper like this one may call on beyond what the body used, so that a change to the engine that breaks any of them shows here first.

B.1 Text

[1] The code is available at `https://example.org/navier-stokes-energy`.

Styles: **bold**, *italic*, *slanted*, SMALL CAPITALS, sans serif, `typewriter`, *emphasis with* emphasis *inside*, and sizes from tiny and small to large and Large. Marked text: highlighted, underlined, ~~struck through~~, underlined again and ~~struck out~~. Characters by name: §, †, ©, ®, €, £, æ, ø, ß, é, à, ô, ü, ñ, ç, š, and the logos TEX and LATEX. A time step is limited by the Courant–Friedrichs–Lewy (CFL) condition, and after that simply by the CFL condition; a temperature reads 20 °C, and a chemical reaction $2H_2 + O_2 \rightarrow 2H_2O$.

B.2 Boxes

Inline boxes keep their text together: framed, shaded, flush right, raised, xy, and one saved and used twice, **saved saved**. The graphicx transforms turn, scale and mirror a box: up, larger, fitted and mirrored.

A minipage on the left, set as a column of its own, its lines as wide as the minipage and no wider.

And one on the right, beside it, its first line level with the first line of its neighbour.

A framed block, which runs across the column with a rule around it.

A titled box

A colored box with a title bar, as tcolorbox sets one.

B.3 Lists

(i) Numbered in lower-case Roman numerals, in parentheses,

(ii) with a nested list inside,

- bulleted,
- and a second item,

(iii) and a third.

B.4 Tables

Table 2 uses a paragraph column, a centred column of a fixed width, a cell over two columns and a colored row; beside this sentence an inline table stands on its first row and another stands on its last .

Table 2: A table of every kind of column.

Left	Centred	Wrapped
one	two	a paragraph column wraps its text to whatever room the table leaves it
three	four	two lines
over two columns		five

B.5 Code

Code is set as written, `a_b = c % d` inline and in blocks:

```
for (int i = 0; i < n; ++i) sum += a[i];   // kept as typed
```

Listing 1: The projection step

```
def project(u, p, dt):
    return u - dt * grad(p)

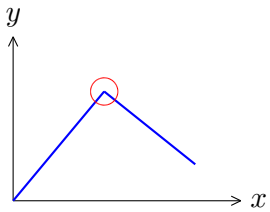
energy = 0.5 * sum(u * u) * h**2
```

Listing 1 is the step of algorithm 1 that enforces incompressibility.

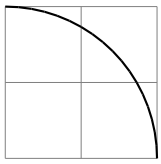
B.6 Columns

Two columns balanced against each other, as the multicol package sets them. The text runs down the first column and on into the second, and the two end level with each other where the block ends, however much text it holds. After the block, the text runs across the whole page again, as this paragraph shows by carrying on for long enough to fill a few lines in each of the two columns it is set in.

B.7 Drawing



(a) A path and a circle.



(b) A grid and an arc.

Figure 2: Two drawings side by side, fig. 2a and fig. 2b.

B.8 The machinery underneath

What a package is written in is here too, and works the way TeX's own does: a count advanced in a loop gives 321, a length multiplied gives 6.0pt, a token list gives kept tokens, e-TeX's expressions give 9 and 3.5pt, and the year is written mmxxvi in Roman numerals. The name \relax prints as itself and SHOUTS in capitals.

C Kernel checks

This appendix sets what LaTeX's own kernel gives a document, with no package asked for: every heading, face, size, accent and special character, the spacing commands, the text blocks, lists four deep, notes, counters, lengths, a document's own commands and blocks, boxes, tables, LaTeX's own picture, and mathematics of every kind.

C.1 Headings

C.1.1 A subsubsection

Text under a subsubsection.

A paragraph heading. Run in, as LaTeX's are.

A subparagraph. Run in and indented.

An unnumbered subsubsection

Text under it.

C.2 Faces and sizes

Families: roman, sans serif, `typewriter`; series: **bold**, medium; shapes: *italic*, *slanted*, SMALL CAPITALS, upright, and the normal face. The same as commands: roman, sans, mono, **bold**, medium, *italic*, *slanted*, CAPS, upright, normal, *emphasised* and *emphasised too*, and underlined. Combined: ***bold italic*** and **bold sans**. Sizes, every one: tiny scriptsize footnotesize small normalsize large Large LARGE huge Huge.

C.3 Characters

Accents on every vowel they take: á é í ó ú, à è ò, â ê ô, ä ë ö ü, ã ñ õ, ā ō, ż, ă, č š ž, ő, ç ş, ạ, a̲, å and o͡o. Letters of other alphabets: æ Æ œ Œ å Å ø Ø ł Ł ß ı ȷ. Specials written as text: # $ % & _ { } \ ~ ^ | < >. Punctuation: "double quotes", 'single quotes', a hyphen-ated word, an en dash 1–2, an em dash—so, an ellipsis… and ..., § 1, ¶, †, ‡, •, ·, ©, ®, ™, £, «guillemets», and / between words. The logos TEX, LATEX and LATEX 2ε, and today, October 20, 2026.

C.4 Spacing

Horizontal space: a b (thin), a b (medium), a b (thick), a b (control space), a b, a b, a b, a b and a negative one, ab; a tie keeps words together. Lines filled: Left Right
Dots ..... there
Rule here

Vertical space: a small skip,

a medium one,

a big one,

and a set one.

Line breaks without a new paragraph:
a second line
and a third;
a break asked for, and a word that may break where marked: inconceivable. Invisible material keeps its room: [ ], [ ], [x] and smashed.

C.5 Text blocks

Centred text, a line at a time.
A second centred line.

Flush left, ragged right.

Flush right, ragged left.

A short quotation, indented on both sides.

> A longer quotation, its paragraphs indented as well as its margins.
>
> Its second paragraph.

A verse, its lines as written,
and turned over when too long
for the measure it is set in.

A centred paragraph by declaration.

A paragraph ragged right by declaration, long enough to show its right edge left uneven.

A paragraph ragged left.

Ada	engines	1843
Alan	machines	1936

C.6 Lists, four deep

1. First level, numbered.
   (a) Second level, lettered.
      i. Third level, in Roman numerals.
         A. Fourth level, in capitals.

⋆ An item with a label of its own.

2. An item referred to later.

- A bullet,
  – a dash,
    * an asterisk,
      · and a dot.

Term its description,

Another term and another, long enough to run on to a second line under the first.

C.7 Notes

A footnote[2], a mark set apart[3] with its text given after, and a note in the margin beside its line. A margin note.

C.8 Counters and lengths

The counter is 8, viii, VIII, h, H and ‡; its value 8, and 8 as printed. A length set to a phrase's width: 88.875pt. The page's own: `\textwidth` is 453.01624pt, `\parindent` 17.0pt, and `\baselineskip` 13.59999pt.

C.9 A document's own commands and blocks

Dear reader, Hello world, **loud!**.
Note: *A block of the document's own.*
Remark: *With its optional argument.*

[2]A note at the foot of the page.

[3]The text of the mark.

Definitions in TeX's own words: (1,2), (3,4) and (a,b). made by name, and the two are the same; three is more; a centimetre is longer; two.

C.10 Boxes

an mbox, left in 3cm , centred , an fbox, right, framed, up and down, a rule ——— and a strut. A parbox, three centimetres wide, its text wrapped within it. Another, standing on its last line. kept and kept.

C.11 Tables

Table 3: LaTeX's own tabular, every rule it has.

A heading over every column			
left	centre	right	a paragraph column wraps its text
a	b	1.5	short
two columns		3	last

no outer space at either side

C.12 LaTeX's picture

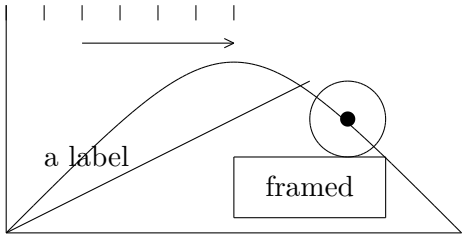


Figure 3: Lines, a vector, circles, a label, a framed box, a row of ticks and a curve, in LaTeX's own picture.

C.13 Mathematics

In the line, $a^2 + b^2 = c^2$, x_i, y, and z outside a formula; displayed:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \qquad \prod_{i=1}^{n} i = n!, \qquad \lim_{x \to 0} \frac{\sin x}{x} = 1.$$

LaTeX's own numbered and aligned equations:

$$\begin{aligned} f(x) &= (x+1)^2 \\ &= x^2 + 2x + 1 \end{aligned} \tag{6}$$

Every Greek letter: $\alpha\beta\gamma\delta\epsilon\varepsilon\zeta\eta\theta\vartheta\iota\kappa\lambda\mu\nu\xi\pi\varpi\rho\varrho\sigma\varsigma\tau\upsilon\phi\varphi\chi\psi\omega$ and $\Gamma\Delta\Theta\Lambda\Xi\Pi\Sigma\Upsilon\Phi\Psi\Omega$. Alphabets: rm *it* **bf** sf tt $\mathcal{CAL}$ $\mathbb{BB}$ $\mathfrak{Frak}$. Accents: $\hat{a}\ \check{a}\ \tilde{a}\ \acute{a}\ \grave{a}\ \dot{a}\ \ddot{a}\ \breve{a}\ \bar{a}\ \vec{a}\ \widehat{abc}\ \widetilde{abc}\ \overline{abc}\ \underline{abc}$. Relations and operators: $\leq\geq\neq\approx\equiv\sim\simeq\cong\propto\subset\supseteq\in\notin\ni\perp\|\|\pm\mp\times\div\cdot\circ\bullet\oplus\otimes\cup\cap\wedge\vee\setminus$. Arrows: $\leftarrow\rightarrow\leftrightarrow\Leftarrow\Rightarrow\Leftrightarrow\mapsto\uparrow\downarrow\longrightarrow\hookrightarrow$. Dots: $1, ..., n$, $1 + \cdots + n$, and in a matrix:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \quad \overbrace{a+b+c}^{\text{three}} \quad \underbrace{x+y}_{\text{two}} \quad \sqrt[3]{x} \quad \sqrt{\frac{a}{b}}$$

$$\stackrel{\text{def}}{=} \quad a \bmod b \quad a \equiv b \pmod{m} \quad \sqrt[2]{p\,q} \quad \overbrace{r\,s}^{2}$$

$$\left\langle \frac{x}{y} \right\rangle \quad \left(\left(x \right) \right) \quad \int_0^\infty \oint \iint \quad \sum_i \sum_j$$

C.14 References

Every kind of reference in one place: appendix C.1 is a heading, equation (6) LaTeX's own, table 3 a table, fig. 3 a picture, item 2 an item, and appendix C starts on page 7.

References

[1] J. T. Beale, T. Kato and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Comm. Math. Phys.*, 94:61–66, 1984.

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